

COTSWOLDS NATIONAL LANDSCAPE

Ecological Permeability & Least-Cost Connectivity Network
Focal Scenario: Woodland-Associated Mammals

LANDSCAPE CONNECTIVITY SUMMARY

- Total Landscape Extent: 2,041 km²
- Ancient Woodland Cores (≥5ha): 400 patches (8,068 ha)
- Functional Corridors: 1,308 least-cost edges
- Disconnected Sub-networks: 82 components
- Isolated Habitat Cores: 41 patches
- Critical Restoration Bottlenecks: 844 pinch points
- Keystone Hub: Core #223 (Betweenness: 0.00293)
- Base Friction Surface: UKCEH LCM2023 (10m) + OS Roads

MAP LEGEND & THEMES

Network Stepping Stones (Betweenness Centrality)

- Local Stepping Stone
- Regional Connector
- Strategic Stepping Stone
- Primary Keystone Core Hub
- Restoration Bottlenecks & Pinch Points
- Ancient Woodland Core Patches (≥ 5 ha)

Least-Cost Corridors (Cost / Euclidean)

- Direct / High Permeability (<1.2)
- Moderate Permeability (1.2-1.6)
- Constrained Path (1.6-2.2)
- Severely Impeded (>2.2)
- Cotswolds National Landscape Boundary



Coordinate Reference: British National Grid (EPSG:27700)
Pipeline: Dijkstra Least-Cost Path on 50m UKCEH Resistance Grid | Network Centrality
Data: UKCEH LCM2023, Natural England (AW/PHI), OS Open Roads, © OpenStreetMap